MTH 202 **Chapter 5: Topics In Vector Calculus**

5.1 Vector Field Worksheet

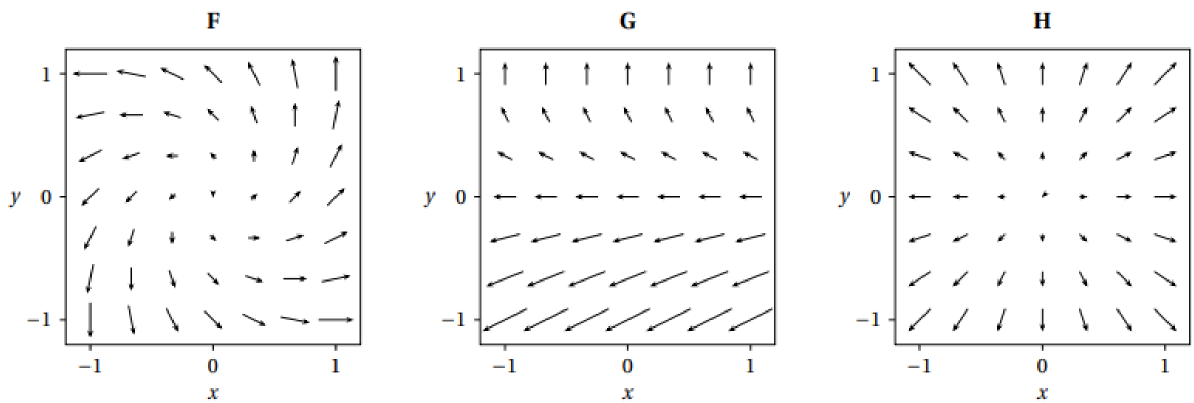
1. Sketch the vector field $\mathbf{F}(x, y) = -y\mathbf{j}$.

2. Let $f(x, y) = x^2 + y^2$. Sketch the vector field $\nabla f(x, y)$

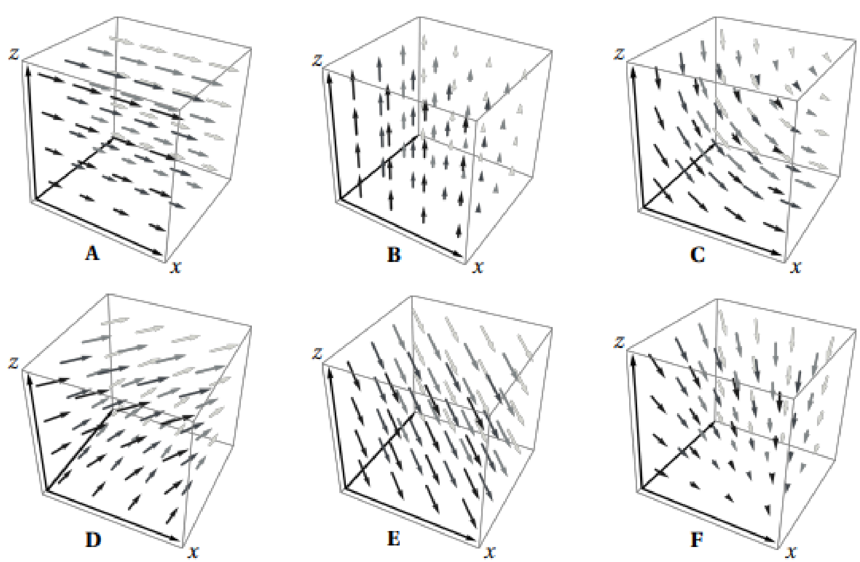
3. Sketch the vector field $\mathbf{F}(x, y) = \langle y, -x \rangle$.

4. Compute the gradient vector field of f, where $f = \arctan\left(\dfrac{x}{y}\right)$

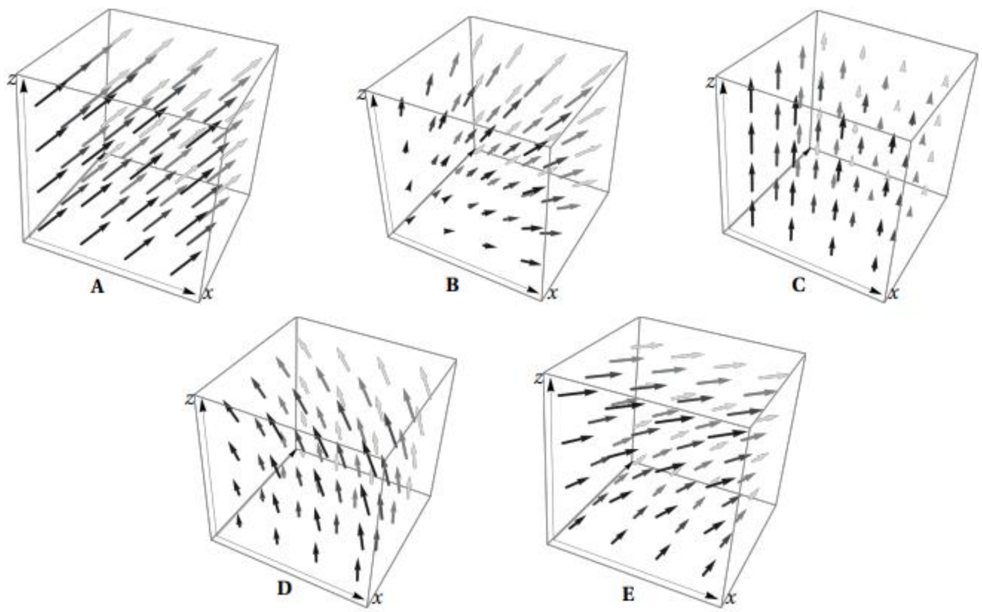
5. Three vector fields are shown below. Which of the vector field corresponds to $\langle y-1, y\rangle$?



6. Here are plots of 6 vectors fields on the box where $0 \le x \le 1$, $0 \le y \le 1$, and $0 \le z \le 1$. Which of the following vector fields corresponds to the vector field $\mathbf{F} = \langle z, 1, 0\rangle$?



7. Here are plots of 5 vectors fields on the box where $0 \le x \le 1$, $0 \le y \le 1$, and $0 \le z \le 1$. Which of the following vector fields corresponds to the vector field $\mathbf{F} = \langle -z, 0, 1+x \rangle$?



8. Determine if the vector field $\mathbf{F}(x, y) = \left\langle \ln(y) + 2xy^3, 3x^2y^2 + \frac{x}{y} \right\rangle$ is conservative.

9. Determine a potential function for the vector field $\mathbf{F}(x, y) = \langle y + 2x, x + 3y^2 + \sin(y) \rangle$.